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Investor Inattention and Friday Earnings Announcements

STEFANO DELLAVIGNA and JOSHUA M. POLLET*

ABSTRACT

Does limited attention among investors affect stock returns? We compare the response to earnings announcements on Friday, when investor inattention is more likely, to the response on other weekdays. If inattention influences stock prices, we should observe less immediate response and more drift for Friday announcements. Indeed, Friday announcements have a 15% lower immediate response and a 70% higher delayed response. A portfolio investing in differential Friday drift earns substantial abnormal returns. In addition, trading volume is 8% lower around Friday announcements. These findings support explanations of post-earnings announcement drift based on underreaction to information caused by limited attention.

INVESTORS HAVE A LIMITED AMOUNT OF TIME AND COGNITIVE RESOURCES to process information. Despite the intuitive appeal of limited attention, little evidence exists on the extent to which the quality of decision-making by investors declines in response to distractions. Incentives, information aggregation across investors, and arbitrageurs may eliminate the effects of limited attention.

We examine a decision in which attention to new information plays a crucial role: the response to earnings surprises. We compare announcements that occur just before the weekend, on Friday, to announcements on other weekdays. If weekends distract investors and lower the quality of decision-making, the immediate response to Friday earnings surprises should be less pronounced. As investors revisit their decisions in subsequent periods, the information should eventually be incorporated in stock prices. As a result, the delayed response, measured by the post-earnings announcement drift, should be of greater

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magnitude for Friday announcements. If instead, limited attention does not affect stock prices, the response to announcements should not differ for Friday and non-Friday announcements.

This paper addresses the debate on the explanation for the post-earnings announcement drift (Bernard and Thomas (1989) and Chan, Jegadeesh, and Lakonishok (1996)). Behavioral explanations depend on the disposition effect (Frazzini (2006) and Grinblatt and Han (2005)), fluctuations in overconfidence (Daniel, Hirshleifer, and Subrahmanyam (1998)), beliefs about mean reversion (Barberis, Shleifer, and Vishny (1998)), or underreaction to information due to cognitive limits (Hong and Stein (1999)). Of the explanations in the literature, only underreaction to information makes the prediction that distractions increase the drift.

The paper proceeds as follows. In Section I, we present a model of the response of stock prices to signals about earnings. In each period, a share of the agents is distracted and does not observe a signal regarding company performance. Given limits to arbitrage in the form of risk aversion, the distracted agents affect prices. A larger share of inattentive investors shrinks the immediate response and magnifies the delayed response of prices to the signals. Therefore, distraction increases the post-earnings announcement drift. The combined response to the announcement, however, is not affected by the distracted investors. We also show that companies that maximize short-term share value release bad news on high distraction days.

In Section II, we characterize our sample of earnings announcements from January 1995 until June 2006. Since we analyze the difference between Friday and non-Friday announcements, the accuracy of the announcement date is critical. We devise a rule, based on newswire announcement dates, that identifies the correct announcement date from I/B/E/S and COMPUSTAT data with more than 95% accuracy. This rule can be used to improve the accuracy of the I/B/E/S announcement date from 1984 onward.

In Section III, we evaluate the immediate and delayed stock return response to information by comparing the top and bottom quantile of the earnings surprise. The immediate stock response is 15% lower for Friday announcements than for non-Friday announcements. Conversely, the delayed response is 70% larger for Friday announcements. We combine these findings in a summary measure. The delayed response as a percentage of the total response is 60% on Friday and 40% on other weekdays. According to our model, these estimates imply that the share of distracted agents in the economy is at least 60% on Friday and 40% on other days.

While our initial analysis is focused on the top and bottom quantiles, we also examine the immediate and delayed response for the entire sample of earnings announcements. The results from this regression analysis are quantitatively similar and more precisely estimated. In addition, we construct portfolios to measure the differential post-earnings announcement drift for Friday announcements. A trading strategy that purchases (long position) the Friday drift and sells (short position) the non-Friday drift earns abnormal returns of approximately four percentage points per month.

These findings are consistent with weekend distractions altering the investor response to earnings information. Individuals are more likely to underreact initially to Friday announcements. Eventually, investors become aware of the information they neglected and trade accordingly. The stronger delayed response (larger drift) reverses the initial underreaction.

If investors are more distracted on Friday, we also expect lower abnormal trading volume for Friday announcements. In Section IV, we find that the abnormal volume is indeed 8% lower for Friday announcements than for non-Friday announcements.

The stock return and volume results suggest that investors exhibit a lower immediate response to Friday announcements. Managers that maximize short-term value should respond by releasing worse announcements on Friday. Hence, the results in this paper provide an explanation for the known findings of worse earnings (and dividend) announcements on Friday (Penman (1987), Damodaran (1989), Bagnoli, Clement, and Watts (2005)).

In Section V, we consider alternative explanations of the findings. First, the slower response to Friday announcements may be due to mechanical differences between Friday and non-Friday announcements due to the extended trading stoppage or to a larger share of announcements made after the market is closed. However, the 2-day break after Friday announcements should allow investors more time to respond and thereby reduce the underreaction. Moreover, during the sample period Friday announcements are *less* likely to be released after the market closes. Second, the slower response to Friday announcements may be caused by pre-announcement leakage of information. However, the stock returns during the 30 days before the announcement do not differ systematically for Friday and non-Friday announcements after conditioning on earnings news. Third, the slower response to Friday announcements could be due to systematic differences in the characteristics of companies announcing on Friday. While it is impossible to fully control for company heterogeneity, the results are largely unaffected by the introduction of time, market capitalization, and earnings surprise volatility controls.

The results in this paper are related to the literature on inattention in finance (Barber and Odean (2008), Cohen and Frazzini (2008), DellaVigna and Pollet (2007), Dyck and Zingales (2003), Hirshleifer and Teoh (2003), Hong and Stein (1999), Huberman and Regev (2001), Peng and Xiong (2006)) and economics (Gabaix et al. (2006)). From this standpoint, our key finding is that a proxy for inattention increases the delayed reaction of stock prices to new information. This result supports the theory that momentum (Jegadeesh and Titman (1993)) and post-earnings announcement drift (Bernard and Thomas (1989) and Chan et al. (1996)) are caused by underreaction to new information due to cognitive constraints. In a related paper, Hirshleifer, Lim, and Teoh (forthcoming) show that the response of stock prices to earnings news is delayed on days with more earnings announcements, a proxy of distraction. The evidence in our paper about the speed of information assimilation by weekday differs from the previous evidence on aggregate return and trading volume by weekday (French (1980), Keim and Stambaugh (1984)).

Finally, this paper is another example of a market response to a bias, in this case due to investor inattention. This is a long-standing theme in finance (Baker, Ruback, and Wurgler (2007), DeLong et al. (1990), Shleifer (2000)) and it has more recently been applied to firm pricing (DellaVigna and Malmendier (2004), Gabaix and Laibson (2006)).

I. Model

We present a model of portfolio choice where all investors are exposed to a signal about next period's dividend, but the fraction of investors that pay attention to this signal varies (e.g., by weekday). While managers cannot manipulate the signal itself as in Hirshleifer and Teoh (2003), they choose to announce the signal when either a high fraction or a low fraction of investors is distracted.

A. Setup

The risky asset has price P_t , pays a risky dividend D_{t+1} before the start of period $t + 1$, and is in fixed supply. The dividend D_{t+1} is equal to $\delta + s_t + \varepsilon_{t+1}$, where $s_t \sim N(0, \sigma_s^2)$ is the signal broadcast to the public before the start of period t and $\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$ is the random component that is unknown until the end of period t . We assume that s_t and ε_t are contemporaneously independent and are drawn identically and independently across periods. The riskless asset has a price normalized to one, a gross return of $1 + R$ every period, and is in perfectly elastic supply.

There is a continuum of investors on the interval $[0, 1]$ where a fraction $1 - \mu_t$ of investors observes the signal s_t (the attentive investors) and a fraction μ_t of investors does not (the distracted investors). The manager observes the signal s_t and then decides to release the signal on a high distraction day, denoted $\mu_t = \mu_h$ (e.g., Friday), or on a low distraction day, denoted $\mu_t = \mu_l$ (e.g., another weekday), where $0 \leq \mu_l \leq \mu_h \leq 1$. The model analyzes an overlapping generations framework where all wealth is consumed at the end of each period (quarter).

B. Portfolio Choice

At the beginning of quarter t , agent i invests λ_t^i units in the risky asset in order to maximize end-of-period wealth W_t^i . Attentive investors use the signal s_t to make this decision, while distracted investors do not. We define $E_t^i[\cdot]$ and $\sigma_{i,t}^2(\cdot)$ as the conditional expectation and variance operators, respectively, using the information set for individual i . For example, the expected dividend for attentive investors (where $i > \mu_t$) is denoted $E_t^i[D_{t+1}] = E_t^{1-\mu}[D_{t+1}] = \delta + s_t$ and the expected dividend for distracted investors (where $i \leq \mu_t$) is denoted $E_t^i[D_{t+1}] = E_t^\mu[D_{t+1}] = \delta$. Each investor has a quadratic utility function with risk aversion parameter $\gamma > 0$. The i^{th} individual solves

$$\begin{aligned} & \max_{\lambda_t^i} E_t^i[W_{t+1}^i] - \frac{\gamma}{2} \text{Var}_t^i[W_{t+1}^i] \\ \text{s.t. } & W_{t+1}^i = \lambda_t^i (P_{t+1} + D_{t+1} - P_t) + (W_t^i - \lambda_t^i P_t)R + W_t^i. \end{aligned} \quad (1)$$

Hence, the demand for the risky asset is

$$\lambda_t^i = \frac{E_t^i[P_{t+1} + D_{t+1}] - P_t(1 + R)}{\gamma \sigma_{i,t,P+D}^2}. \quad (2)$$

C. Equilibrium

We impose the market clearing and transversality conditions to solve for the equilibrium price:

$$P_t = \frac{\delta - \bar{a}}{R(1 + R)} + \frac{\delta - a_t}{1 + R} + \frac{1 - b_t}{1 + R} s_t, \quad (3)$$

where $a_t = [\mu_t/(\gamma \sigma_{\mu,t,P+D}^2) + (1 - \mu_t)/(\gamma \sigma_{1-\mu,t,P+D}^2)]^{-1}$, $b_t = \mu_t/(\gamma \sigma_{\mu,t,P+D}^2) a_t$, $\bar{a} = \eta a(\mu_h) + (1 - \eta) a(\mu_l)$, and $\eta = \text{prob}(\mu_{t+s} = \mu_h)$. Since b_t increases with the fraction of distracted agents, inattention causes prices to be less responsive to signals.

D. Response to Signal

We derive measures of the immediate and the delayed response of the stock price to the signal (e.g., an earnings announcement). Define the dollar excess return from $t - 1$ to t as $Z_t = P_t + D_t - (1 + R)P_{t-1}$. The abnormal return from $t - 1$ to t is the change in expected returns due to new information. This abnormal return incorporates two sources of new information: the signal s_t and the unexpected dividend $D_t - E_{t-1}[D_t]$. Since the second component is unrelated to the signal, we define the immediate response to the signal as $E_t[IR_t] \equiv Z_t - E_{t-1}[Z_t] - (D_t - E_{t-1}[D_t])$.

The excess return for the second period (t to $t + 1$) is Z_{t+1} . The change in the expectation for this excess return caused by the signal is $E_t[DR_{t+1}] \equiv E_t[Z_{t+1}] - E_{t-1}[Z_{t+1}]$. This measure captures the delayed response and is the theoretical equivalent of the post-earnings announcement drift. We also define the long-term response to the announcement $E_t[LR_{t+1}]$ as the discounted sum of $E_t[IR_t]$ and $E_t[DR_{t+1}]$. Once D_{t+1} is paid to investors, the information in the signal s_t is irrelevant for all subsequent investment decisions because expectations of future cash flows do not depend on s_t . Hence, the impact of distraction at t on abnormal returns is fully resolved by $t + 1$.¹

PROPOSITION 1: (i) *The immediate response $E_t[IR_t]$ is a linear function of the signal s_t , with slope coefficient $(1 - b_t)/(1 + R)$. The slope $(1 - b_t)/(1 + R)$ is decreasing in the share of distracted investors μ_t and is equal to $1/(1 + R)$ if $\mu_t = 0$.* (ii) *The delayed response $E_t[DR_{t+1}]$ is a linear function of the signal s_t , with slope*

¹ Given the information and trading structure of the model, the behavior of the stock return remains unchanged if we assume that the distracted investors observe the signal one period after announcement or that the distracted investors never observe the signal. This result is driven by the inability to trade between the announcement of the signal at t and the dividend payment at $t + 1$, when the information in the signal becomes irrelevant.

coefficient b_t . The slope b_t is increasing in the share of distracted investors μ_t and is equal to zero if $\mu_t = 0$. (iii) The long-term response $E_t[LR_{t+1}]$ is a linear function of the signal s_t , with slope coefficient $1/(1 + R)$. The slope $1/(1 + R)$ is independent of the share of distracted investors μ_t .

Inattentive agents delay the incorporation of information into the stock price. A larger fraction of these distracted agents lowers the immediate response and increases the delayed response to information. The immediate response on a high distraction day is approximately proportional to the immediate response on a low distraction day for identical signals, with the proportionality factor $(1 - b_h)/(1 - b_l)$. The long-term response does not depend on the share of inattentive investors.

We consider a summary measure for the delayed incorporation of news into stock prices. Let s_A and s_B be signals of different quality, where $s_A > s_B$. We define the delayed response ratio as the ratio of the differential delayed response to the differential long-term response: $DRR_t = (E[DR_t | s_A, \mu_t] - E[DR_t | s_B, \mu_t]) / (E[LR_t | s_A, \mu_t] - E[LR_t | s_B, \mu_t])$.

COROLLARY 1: (i) The delayed response ratio DRR_t equals $b_t(1 + R)$. (ii) DRR_t is increasing in the share of distracted investors μ_t and is equal to zero if $\mu_t = 0$. (iii) DRR_t provides a bound on the share of distracted investors: $\mu_t \geq DRR_t$.

On high distraction days a greater fraction of the long-term response is delayed. If there are no distracted investors, the delayed response ratio is zero. We use the bound $\mu_t \geq (1 + R)DRR_t$ in Section III.

Manager optimization. The manager of the firm can announce the signal on a high distraction day ($\mu_t = \mu_h$) or a low distraction day ($\mu_t = \mu_l$). Long-term managers maximize the expected long-term price $E_t[P_{t+1}]$ and short-term managers maximize the current price $E_t[P_t] = P_t$.

PROPOSITION 2: (i) Managers that maximize long-term value are indifferent between μ_l and μ_h . (ii) Managers that maximize short-term value choose $\mu_t = \mu_h$ whenever $s_t \leq -\sigma_s^2 \gamma$. It follows that $E[s_t | \mu_t = \mu_h] < E[s_t | \mu_t = \mu_l]$ and $E[IR_t | \mu_t = \mu_h] < E[IR_t | \mu_t = \mu_l]$.

Since the long-term price is independent of the fraction of distracted investors, long-term managers are indifferent about the release decision. Short-term managers with sufficiently negative news choose to release the signal on high distraction days because they prefer an attenuated immediate response. The threshold rule used by short-term managers implies that the average signal and immediate response are worse on high distraction days. Hence, investor inattention explains the well-documented finding that earnings surprises and returns surrounding earnings announcements are lower for Friday announcements. (Penman (1987), Damodaran (1989), Bagnoli et al. (2005)).

II. Data and Summary Statistics

A. Data

Our sources of earnings data are I/B/E/S and COMPUSTAT. We begin with all quarterly earnings announcements from I/B/E/S for which at least one analyst forms an earnings forecast in the 90 days before the announcement. We restrict the sample to announcements that have stock return data in CRSP and are reported in both I/B/E/S and COMPUSTAT with a difference of at most five calendar days between the reported announcement dates. The resulting sample includes 228,651 announcements from January 1984 to June 2006.

We construct a measure of the announcement date using the reported COMPUSTAT and I/B/E/S dates. In order to quantify the accuracy of these dates, we randomly select 2,487 earnings announcements for the period 1984 to 2002 and use Lexis-Nexis to search the announcement date in the PR newswires. We look for the function of the COMPUSTAT and I/B/E/S dates that matches the newswire date most accurately. We oversample announcements that occur on Friday according to I/B/E/S. The results of the search (available upon request) suggest that the optimal imputation rule for the date differs for three categories of announcements:

1. I/B/E/S and COMPUSTAT announcement dates differ. In the case of disagreement, the earlier date is usually the actual date of the announcement, and the later date is the date of publication in the *Wall Street Journal*. We impute the date to be the earlier one.
2. Before January 1, 1990: I/B/E/S and COMPUSTAT announcement dates agree. In this case, most announcements are recorded using the *Wall Street Journal* date in both data sets. We assign the announcement date to be the previous trading date.
3. After January 1, 1990: I/B/E/S and COMPUSTAT announcement dates agree. During this time period, the announcement date is usually from a newswire source. We impute the date to be the I/B/E/S and COMPUSTAT date.

After applying these rules, we measure the accuracy of the imputed announcement date (Table A.I in the Appendix) for Friday and non-Friday announcements. Before 1995, a high number of earnings announcements were recorded with an error of at least 1 trading day. In addition, the errors are more common for Friday announcements than for non-Friday announcements. During the more recent years, the accuracy of the earnings date has increased substantially, and is almost perfect after December 1994. For the purposes of this paper, even a 1-day error in the date is important, since it would lead to a misclassification of Friday announcements. In light of this evidence, we limit the analysis to the 154,051 earnings announcements taking place after 1994. In this period, the imputed earnings announcement date is correct more than 95% of the time for both Friday and non-Friday announcements.

We use the consensus analyst forecast from I/B/E/S as a proxy for the expectations of investors. The consensus forecast is defined as the median forecast

among all the analysts that make a forecast in the last 30 calendar days before the earning announcement. If an analyst has made multiple forecasts in this time horizon, we use only the most recent one.²

We define the earnings surprise as the difference between the earnings announcement and the consensus earnings forecast, normalized by the price of a share (Kothari (2001)). Let $e_{t,k}$ be the earnings per share announced in quarter t for company k and $\hat{e}_{t,k}$ be the corresponding consensus analyst forecast.³ Further, indicate by $P_{t,k}$ the price of the shares of company k 5 trading days before the announcement in quarter t . The earnings surprise $s_{t,k}$ is

$$s_{t,k} = \frac{e_{t,k} - \hat{e}_{t,k}}{P_{t,k}}. \quad (4)$$

We match the announcement dates with information on stock returns and trading volume from CRSP. We construct cumulative abnormal returns for different windows around the announcement date. Let $R_{u,k}$ be the stock return of company k on day u and let $R_{u,m}$ be the market return on day u . We obtain $\hat{\beta}$ for company k in quarter t from the regression $R_{u,k} = \alpha_{t,k} + \beta_{t,k}R_{u,m}$ for days u from $\tau - 300$ to $\tau - 46$, where τ is the date of the announcement in quarter t . The buy-and-hold abnormal return $R_{t,k}^{(h,H)}$ over time period $(\tau + h, \tau + H)$ for stock k in quarter t is then computed as $[\Pi_{j=\tau+h}^{\tau+H}(1 + R_{j,k})] - 1 - \hat{\beta}_{t,k}[\Pi_{j=\tau+h}^{\tau+H}(1 + R_{j,m}) - 1]$.⁴

We drop observations with a missing earnings surprise (8,853 observations), or in which the earnings announcement $e_{t,k}$ or the earnings forecast $\hat{e}_{t,k}$ is larger in absolute value than the price of a share $P_{t,k}$ (215 observations). We also eliminate penny stocks (812 observations) as well as announcements on Saturday, Sunday, or holidays (306 observations). Finally, we exclude observations with returns in the top and bottom 5/10,000th of the distributions for $R_{t,k}^{(0,1)}$ or $R_{t,k}^{(2,75)}$ (282 observations). The final sample includes 143,583 observations.

B. Summary Statistics

More than 80% of announcements occur on Tuesday, Wednesday, or Thursday, 13.8% occur on Monday, and only 5.7% are on Friday (Table I.A). The model can explain the small share of announcements on Friday—firms that commit to an announcement schedule never commit to announcements on a high-inattention day (Friday).

In Table I.B we present summary statistics for the 8,166 Friday announcements (column 1), and for the 135,417 non-Friday announcements (column 2).

² The results are similar if we use the average forecast as a measure of consensus forecast or if we use analyst forecasts made within a shorter time horizon (15 days).

³ The measure of earnings per share in I/B/E/S reflects capital structure changes. In order to make the units of the earning announcements and forecasts comparable with the units of the price data $P_{t,k}$, we apply the adjustment provided by I/B/E/S. Since the adjustment factor is stored as a truncated number, the resulting variables $e_{t,k}$ and $\hat{e}_{t,k}$ have fractional cents. We round the earnings per share measure $e_{t,k}$ to the nearest cent and the earnings forecast $\hat{e}_{t,k}$ to the nearest half cent.

⁴ The results in the paper are similar if we use raw returns or net returns.

Table I
Summary Statistics

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. The earnings surprise for a quarterly announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Announcements made on Saturday or Sunday are excluded from the sample. Panel A presents the distribution of earnings announcements by weekday. Columns 1 to 2 and 4 to 5 of Panel B present summary statistics with standard deviations in parentheses. Column 3 of Panel B presents the difference between columns 1 and 2, column 6 of Panel B presents the difference between columns 4 and 5, and the standard errors for these differences are in parentheses (* significant at 10%, ** significant at 5%, *** significant at 1%). The Homogeneous Sample refers to announcements by companies that have announcements on Friday and other weekdays at least 10% of the time. Panel C presents the average earnings surprise by quantile for announcements made on Friday and on other weekdays. Quantiles 1 to 5 contain announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain announcements for five quintiles of positive earnings surprises. Quantile 6 contains announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are unequal, the number of observations in each quantile differs. The breakpoints for the quantiles are determined separately for each year.

Panel A: Distribution of Earnings Announcements by Day of the Week						
	All	Monday	Tuesday	Wednesday	Thursday	Friday
Number	143,583	19,861	36,358	36,822	42,376	8,166
Fraction	1.0000	0.1383	0.2532	0.2565	0.2951	0.0569

(continued)

Table I—Continued

Panel B: Differences between Announcements on Friday and Other Weekdays						
	Baseline Sample			Homogeneous Sample		
	Friday (1)	Non-Friday (2)	Difference (3)	Friday (4)	Non-Friday (5)	Difference (6)
Earnings surprise	−0.0039 (0.0353)	−0.0009 (0.0231)	−0.0030 (0.0004)***	−0.0043 (0.0370)	−0.0016 (0.0256)	−0.0027 (0.0005)***
Market Cap (\$M)	3.295 (17,599)	3.564 (15,264)	−269 (199)	3.270 (18,717)	2.630 (14,755)	639 (264)**
Year	1999.52 (3.36)	2000.05 (3.29)	−0.53 (0.0382)***	1999.49 (3.34)	1999.65 (3.29)	−0.16 (.0489)***
Month 1 in Quarter	0.5797 (0.4936)	0.6059 (0.4886)	−0.0262 (0.0056)***	0.5906 (0.4918)	0.5909 (0.4917)	−0.0003 (0.0072)
Month 2 in Quarter	0.3166 (0.4652)	0.3114 (0.4631)	0.0052 (0.0052)	0.3146 (0.4644)	0.3174 (0.4655)	−0.0028 (0.0068)
Month 3 in Quarter	0.1037 (0.3049)	0.0827 (0.2754)	0.0210 (0.0034)***	0.0948 (0.2930)	0.0917 (0.2886)	0.0031 (0.0042)
N	N = 8,166	N = 135,417	N = 143,583	N = 5,896	N = 21,234	N = 27,130

Panel C: Average Surprise by Earning Surprise Quantile										
Quantile	6						8	9	10	11 High
	1 Low	2	3	4	5	6 Surprise = 0				
Friday	−0.0523 833	−0.0065 763	−0.0025 627	−0.0011 585	−0.0004 480	0.0000 995	0.0003 628	0.0007 686	0.0014 737	0.0152 986
Other Days	−0.0454 7,595	−0.0064 7,582	−0.0025 7,587	−0.0011 7,589	−0.0004 7,576	0.0000 20,109	0.0003 15,484	0.0007 15,473	0.0014 15,475	0.0143 15,472
N										

Firms announcing on Friday have more negative earnings surprises and 10% smaller market capitalization. Friday announcements are more prevalent toward the beginning of the sample period. For non-Friday announcements, 60.59% of announcements occur in the first month of the quarter (January, April, July, and October), 31.14% in the second month, and only 8.27% in the third month. Friday announcements are more likely to occur in the second or third month.

Columns 4 and 5 present the summary statistics for companies with 10% to 90% of their announcements on Friday. This criterion excludes companies that rarely announce on Friday (116,251 observations) or that almost always announce on Friday (202 observations). For the remaining sample of 27,130 announcements, the summary statistics for announcements on Friday (column 4) and other weekdays (column 5) are less dissimilar. The controls for month of announcement are no longer significantly different, and the difference in the average year of announcement is reduced to less than 2 months. This subsample, labelled Homogeneous Sample, addresses the concern that companies announcing on Friday may also have unobservable features that differ from companies announcing on other days.

III. Stock Return Response

In this section, we examine the responsiveness of stock returns to earnings surprises at various horizons. We compare the responsiveness for announcements on Friday to the responsiveness for announcements on other weekdays. If the fraction of distracted investors is higher on Friday, there should be less immediate response to Friday announcements, followed by more delayed response; the total long-term response should be unchanged (Proposition 1). The fraction of the stock reaction occurring with delay should be higher for Friday announcements (Corollary 1). We present graphical evidence, followed by an analysis of earnings announcements in the top and bottom earnings quantiles, including delayed response ratios, and then by regressions using all announcements. Finally, we analyze the portfolio returns for a strategy that exploits the difference in drift between announcements on Friday and other weekdays.

A. Graphical Evidence

To measure announcement quality, we divide announcements into 11 bins, ordered by the earnings surprise $s_{t,k}$. Negative earning surprises are in quantiles 1 through 5, followed by zero surprises (quantile 6), and positive surprises (quantiles 7 through 11). The thresholds for the bins are set separately for each year to guarantee an equal number of non-Friday announcements for quantiles 1 through 5 and for quantiles 7 through 11. Since positive surprises are twice as common as negative surprises, quantiles 7 through 11 have twice as many observations as quantiles 1 through 5. Within each bin, we separate the Friday announcements from the non-Friday announcements. While this procedure has the advantage that it separately identifies the 10% to 17% of announcements

with no earnings surprise in each year, we also consider the more traditional decile sorting procedure in the statistical analysis in the next subsection. The threshold for the bins is set each year to sort the non-Friday announcements into 10 equal groups. Since approximately 15% of the announcements are for zero surprises and these announcements fall into either decile 3 or 4, deciles 3 through 5 are of unequal size.

Table I.C reports the average earnings surprise $s_{t,k}$ within each quantile for the Friday and the non-Friday announcements. The within-quantile average earnings surprise is very similar for Friday and non-Friday announcements. The only exception is quantile 1, where the average surprise for Friday announcements is 10% lower than for non-Friday announcements.

A.1. Immediate Response

In Figure 1(a) we display the immediate response of stock returns, $R_{t,k}^{(0,1)}$, to earnings surprises for Friday and non-Friday announcements. By construction, $R_{t,k}^{(0,1)}$ is the return from the close on the trading day before the earnings announcement to the close on the trading day after the earnings announcement for stock k in quarter t . This measure captures the short-term response to announcements made during trading hours and announcements made after the market is closed.⁵ Compared to announcements on other days, the responsiveness of stock prices to earning surprises is substantially flatter for Friday announcements. Interestingly, most of the underreaction occurs for positive announcements: Companies with positive announcements on Friday are significantly penalized initially. However, even for negative announcements there is less initial reaction on Friday: Stock returns for quantiles 1 through 5 are less negative for Friday than for non-Friday announcements

A.2. Delayed Response

In Figure 1(b) we display the delayed reaction of stock returns, $R_{t,k}^{(2,75)}$, to the earnings announcements. In the non-Friday sample, positive earnings surprises are followed by positive returns in the period subsequent to the announcement, and the pattern is increasing in the magnitude of the surprise. Negative surprises are followed by negative delayed returns, but the magnitudes are smaller. Compared to non-Friday announcements, Friday announcements exhibit more delayed response. The pattern is strongest for the the most negative surprise, but is also present for the positive surprises.

B. Top and Bottom Quantiles

To quantify the graphical findings, we examine the stock response to very positive earnings news (top quantile) and very negative news (bottom quantile).

⁵ Since the time stamp for each announcement is not available in I/B/E/S or COMPUSTAT, we cannot separate announcements made during trading hours from those made after close.

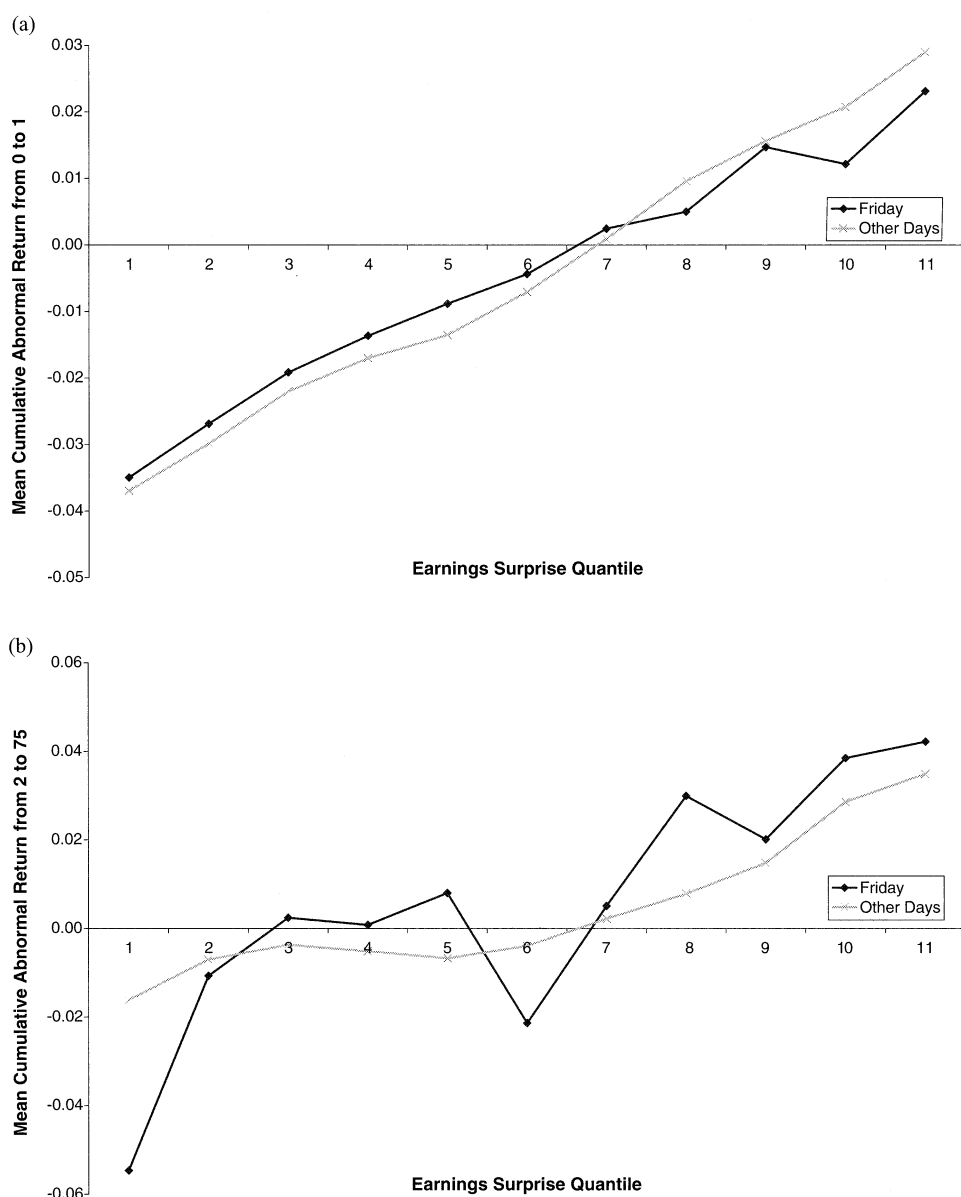


Figure 1a and b. Immediate response to earnings surprises and delayed response to earnings surprises. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year. Announcements made on Saturday or Sunday are excluded from the sample.

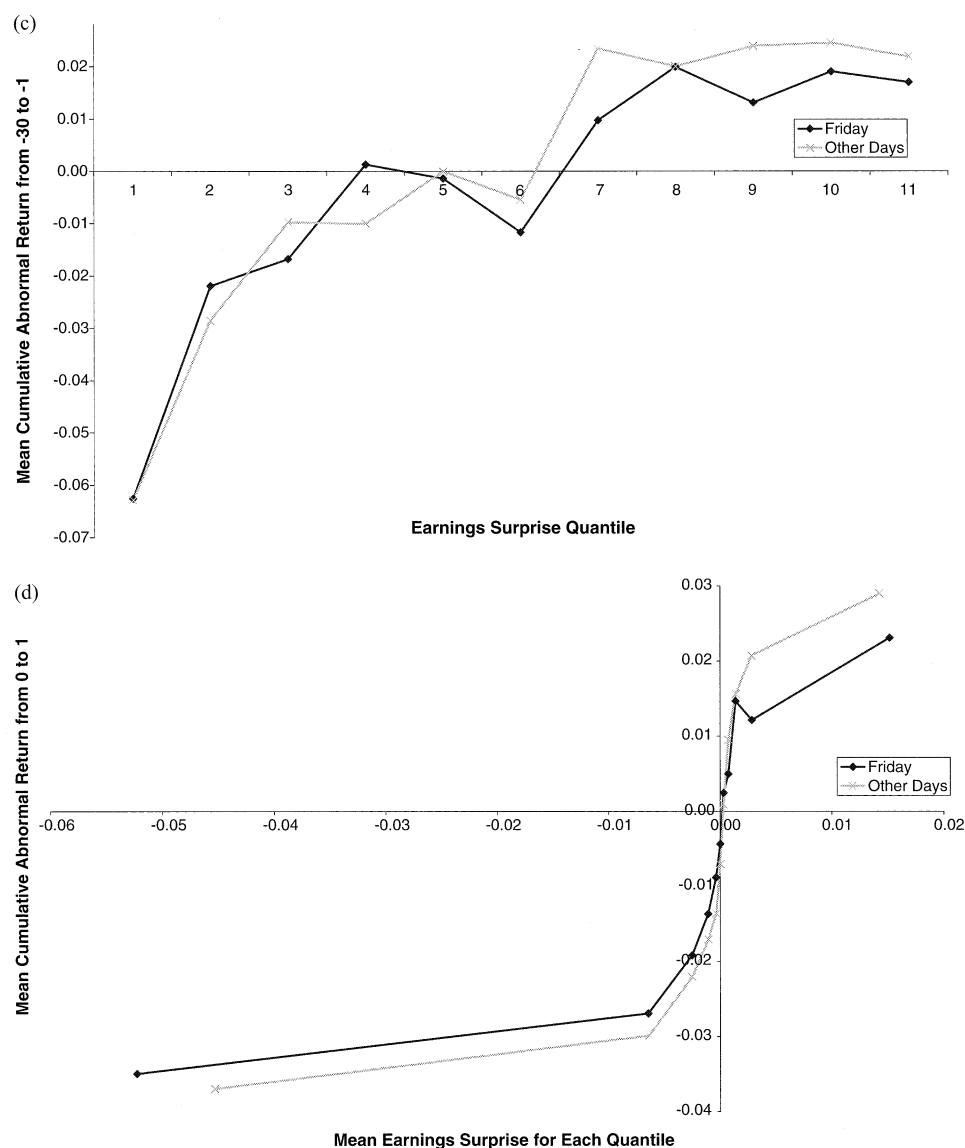


Figure 1c and d. Pre-announcement behavior before earnings surprises and nonlinear form of the immediate response to earnings surprises. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year. Announcements made on Saturday or Sunday are excluded from the sample.

We compare the immediate, delayed, and long-term sensitivity for Friday and non-Friday announcements. The OLS specification is

$$R_{t,k}^{(h,H)} = \phi_B + \phi_{T-B}d_{t,k}^{top} + \phi_B^F d_{t,k}^F + \phi_{T-B}^F d_{t,k}^{top} d_{t,k}^F + \Gamma_0 X_{t,k} + \Gamma_1 d_{t,k}^{top} X_{t,k} + \varepsilon_{t,k}, \quad (5)$$

where $R_{t,k}^{(h,H)}$ denotes the abnormal stock returns for company k in quarter t between h days before the announcement and H days after the announcement. The sample includes only observations in the top quantile(s) ($d_{t,k}^{top} = 1$) or the bottom quantile(s) ($d_{t,k}^{top} = 0$). We focus on the coefficients ϕ_{T-B} and ϕ_{T-B}^F . The coefficient ϕ_{T-B} measures the return to good news (top quantile) relative to bad news (bottom quantile) for non-Friday announcements. The coefficient ϕ_{T-B}^F captures the differential reaction for Friday relative to non-Friday announcements. Under the null hypothesis of constant (or no) investor distraction, ϕ_{T-B}^F should equal zero. Under the alternative hypothesis of higher distraction on Friday, ϕ_{T-B}^F should be negative for the immediate response ($R_{t,k}^{(0,1)}$) and positive for the delayed response ($R_{t,k}^{(2,75)}$). Distracted investors react less to news initially, and more afterwards.

Specification (5) allows the stock response to depend on a set of control variables $X_{t,k}$. For example, the responsiveness of stocks to earnings news may be correlated with company size if profit shocks are more permanent for larger firms. Similarly, the responsiveness may have increased over time if earnings disclosure regulation decreased the pre-announcement leakage of information. We include indicators for year of announcement, as well as indicators for market capitalization deciles. The indicators for market capitalization are constructed from the log market capitalization of company k in quarter t minus the average of the log market capitalization for other companies announcing earnings in the same quarter. We also include month indicators to control for differences in return sensitivity across quarters and within a quarter (early versus late announcements). Finally, in some specifications we also control for earnings surprise volatility. Each year, we sort announcements into deciles using the company's earnings surprise standard deviation during the previous four years, requiring a minimum of four observations. Each control (if included) is also interacted with the indicator variable for the top quantile (or top two quantiles). Standard errors are clustered by day of announcement to control for correlation of returns on the same day.

B.1. Immediate Response

Table II.A presents specification (5) with the immediate abnormal return $R_{t,k}^{(0,1)}$ as dependent variable. Without controls (column 1), the top-to-bottom average return for non-Friday announcements is 6.59% ($\hat{\phi}_{T-B} = 0.0659$). Compared to this value, the top-to-bottom return for a Friday announcement is (marginally) significantly smaller by 0.88 percentage points ($\hat{\phi}_{T-B}^F = -0.0088$). The top-minus-bottom return differential for Friday announcements is not significant with the standard set of controls ($\hat{\phi}_{T-B}^F = -0.0059$, column 2) and

Table II
Stock Price Response to Earnings Announcements for the Top and Bottom Groups

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are unequal, the number of observations in each quantile differ. In some specifications we follow a similar procedure but sort announcements into surprise deciles instead of quantiles.

The breakpoints for the quantiles (deciles) are determined separately for each year. The coefficients in the table are evaluated at the average value of the controls. Columns 1, 2, and 3 analyze observations from the top quantile (11) and the bottom quantile (1). Column 4 analyzes observations from the top decile (10) and bottom decile (1). Columns 5 and 6 analyze observations from the top two quantiles (10 and 11) or the bottom two quantiles (1 and 2). Column 7 analyzes observations from the top two deciles (9 and 10) and bottom two deciles (1 and 2).

The standard set of controls includes indicators for the year of announcement, the month of announcement, and the decile of a firm's market capitalization. The earnings surprise volatility controls in columns 3, 4, 6, and 7 are indicators for the decile of the company's earnings surprise standard deviation during the previous 4 years (requiring at least four observations). If a control is included in a regression, that control is also interacted with the indicator variable for the top quantile (or top two quantiles). Announcements made on Saturday or Sunday are excluded from the sample. Standard errors adjusted for heteroskedasticity and clustered by day of announcement are in parentheses (* significant at 10%, ** significant at 5%, *** significant at 1%).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Constant	-0.0369 (0.0013)***	-0.0370 (0.0024)***	-0.0340 (0.0019)***	-0.0330 (0.0011)***	-0.0334 (0.0009)***	-0.0322 (0.0012)***	-0.0254 (0.0007)***
Friday	0.0021 (0.0040)	0.0003 (0.0040)	0.0015 (0.0043)	0.0010 (0.0031)	0.0024 (0.0027)	0.0012 (0.0029)	0.0008 (0.0022)
Top Group	0.0659 (0.0014)***	0.0634 (0.0015)***	0.0640 (0.0021)***	0.0635 (0.0014)***			
(Top Group) * Friday	-0.0081 (0.0047)*	-0.0059 (0.0048)	-0.0090 (0.0050)*	-0.0071 (0.0044)			
Top Two Groups					0.0582 (0.0010)***	0.0571 (0.0013)***	0.0518 (0.0009)***
(Top Two Groups) * Friday					-0.0092 (0.0031)***	-0.0086 (0.0034)***	-0.0095 (0.0028)***
Standard Controls (Interacted)		X	X	X		X	X
Surprise Volatility Controls (Interacted)			X	X		X	X
Sorting Procedure	11 Quantiles	11 Quantiles	11 Quantiles	10 Deciles	11 Quantiles	11 Quantiles	10 Deciles
R ²	0.0915	0.0990	0.1049	0.1148	0.0864	0.0985	0.0986
N	N = 24,874	N = 24,874	N = 19,888	N = 23,458	N = 49,538	N = 40,331	N = 47,879

(continued)

Table II—Continued

Panel B: The Dependent Variable Is the Cumulative Abnormal Return in Event Time from 2 to 75						
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−0.0165 (0.0057)***	−0.0180 (0.0059)***	−0.0014 (0.0063)	−0.0048 (0.0044)***	−0.0116 (0.0041)***	−0.0029 (0.0041)***
Friday	−0.0382 (0.0148)***	−0.0272 (0.0138)**	−0.0432 (0.0147)***	−0.0258 (0.0107)**	−0.0220 (0.0102)**	−0.0140 (0.0099)***
Top Group	0.0514 (0.0055)***	0.0552 (0.0059)***	0.0484 (0.0070)***	0.0510 (0.0056)***		
(Top Group) * Friday	0.0462 (0.0175)***	0.0392 (0.0171)**	0.0462 (0.0184)**	0.0208 (0.0157)		
Top Two Groups					0.0434 (0.0036)***	0.0453 (0.0044)***
(Top Two Groups) * Friday					0.0305 (0.0118)***	0.0216 (0.0125)***
Standard Controls (Interacted)		X	X	X	X	X
Surprise Volatility Controls (Interacted)			X	X		
Sorting Procedure	11 Quantiles	11 Quantiles	11 Quantiles	10 Deciles	11 Quantiles	11 Quantiles
R ²	0.0056	0.0405	0.0426	0.0401	0.0045	0.0350
N	N = 24,874	N = 24,874	N = 19,888	N = 23,458	N = 49,538	N = 40,331

(continued)

Table II—Continued

Panel C: The Dependent Variable Is the Cumulative Abnormal Return in Event Time from 0 to 75						
	(1)	(2)	(3)	(4)	(5)	(7)
Constant	−0.0527 (0.0057)***	−0.0528 (0.0059)***	−0.0372 (0.0063)***	−0.0382 (0.0045)***	−0.0447 (0.0041)***	−0.0273 (0.0031)***
Friday	−0.0370 (0.0148)**	−0.0278 (0.0138)**	−0.0445 (0.0141)***	−0.0268 (0.0107)**	−0.0204 (0.0104)*	−0.0138 (0.0074)*
Top Group	0.1210 (0.0057)***	0.1218 (0.0056)***	0.1178 (0.0075)***	0.1189 (0.0060)***		
(Top Group) * Friday	0.0368 (0.0182)**	0.0323 (0.0179)*	0.0382 (0.0184)**	0.0136 (0.0161)		
Top Two Groups					0.1046 (0.0037)***	0.1008 (0.0039)***
(Top Two Groups) * Friday					0.0213 (0.0124)*	0.0119 (0.0119)
Standard Controls (Interacted)		X	X	X		X
Surprise Volatility Controls (Interacted)			X	X		X
Sorting Procedure	11 Quantiles	11 Quantiles	11 Quantiles	10 Deciles	11 Quantiles	10 Deciles
R ²	0.0231	0.0564	0.0598	0.0587	0.0206	0.0487
N	N = 24,874	N = 24,874	N = 19,888	N = 23,458	N = 49,538	N = 47,879

Panel D: Ratio of the Delayed Stock Response (2 to 75) to the Long-Term Stock Response (0 to 75)						
	(1)	(2)	(3)	(4)	(5)	(7)
Response ratio for Friday announcements	0.6187 (0.0443)	0.6083 (0.0459)	0.6020 (0.0536)	0.5388 (0.0578)	0.5869 (0.0401)	0.5883 (0.0428)
Response ratio for announcements on other days	0.4244 (0.0270)	0.4491 (0.0273)	0.4068 (0.0361)	0.4257 (0.0270)	0.4144 (0.0207)	0.4444 (0.0200)
Difference between the response ratio for Friday and other days	0.1942 (0.0519)***	0.1591 (0.0517)***	0.1952 (0.0612)***	0.1131 (0.0619)*	0.1725 (0.0451)***	0.1439 (0.0460)***
Standard Controls (Interacted)		X	X	X		X
Surprise Volatility Controls (Interacted)			X	X		X
Sorting Procedure	11 Quantiles	11 Quantiles	11 Quantiles	10 Deciles	11 Quantiles	10 Deciles
N	N = 24,874	N = 24,874	N = 19,888	N = 23,458	N = 49,538	N = 47,879

marginally significant with additional controls for earnings surprise volatility ($\hat{\phi}_{T-B}^F = -0.0090$, column 3). Using deciles to form the outermost groups, the coefficient ϕ_{T-B}^F is negative ($\hat{\phi}_{T-B}^F = -0.0071$, column 4), but not significant.

In columns 5 to 7 we consider specification (5) for the observations in the top two and bottom two quantiles. (The variable $d_{t,k}^{top}$ is now an indicator for the top two quantiles.) The larger sample increases the precision of the estimates substantially. The top-to-bottom return differential for non-Friday announcements is 5.82% ($\hat{\phi}_{T-B} = 0.0582$, column 5). Relative to this differential, Friday announcements are associated with a 15.8% ($0.0092/0.0582$) lower immediate return response, an economically and statistically significant difference. This difference is of about the same size and significance level with the standard set of controls (not shown) and with the additional volatility controls ($\hat{\phi}_{T-B}^F = -0.0086$, column 6). The result is slightly larger if we use deciles to form the top and bottom two groups ($\hat{\phi}_{T-B}^F = -0.0095$, column 7).

B.2. Delayed Response

In Table II.B we estimate specification (5) with the delayed abnormal performance $R_{t,k}^{(2,75)}$ as dependent variable. In the specification without controls (column 1), the estimated post-earnings announcement drift for non-Friday announcements, $\hat{\phi}_{T-B}$, is 5.14%. (We measure drift as the difference in performance between the top and bottom earnings surprise quantiles). This magnitude is consistent with previous estimates (Bernard and Thomas (1990)). The drift for Friday announcements is significantly higher by 4.62 percentage points ($\hat{\phi}_{T-B}^F = 0.0462$), an 89.8% increase compared to other weekdays. The differential drift remains large and significant with both sets of controls (columns 2 and 3) but is somewhat smaller and insignificant using deciles ($\hat{\phi}_{T-B}^F = 0.0208$, column 4). The smaller differential drift obtained using deciles is consistent with Figure 1(b), because the decile approach selects less extreme announcements in the bottom group, which is responsible for much of the differential drift.⁶

In columns 5 to 7 we replicate these results using the top two and bottom two groups. The drift is significantly larger on Friday both without controls (column 5) and with controls (column 6). In this latter specification, the Friday drift is 77.2% ($0.0350/0.0453$) larger than drift on other weekdays. The differential drift on Friday is marginally significant using deciles to form the outermost groups ($\hat{\phi}_{T-B}^F = 0.0216$, column 7).

To better understand the nature of this differential drift, we examine drift at various horizons for announcements on Friday and other weekdays in Figure 2. We measure drift as the difference in the average buy-and-hold abnormal returns between the top and bottom quantile. Most of the differential drift between Friday and non-Friday announcements does not emerge in the first 10 trading days following an announcement. Instead, differential drift becomes more readily apparent approximately 30 trading days after the announcement

⁶ The bottom quantile contains 8,428 observations, while the bottom decile contains 15,045 observations.

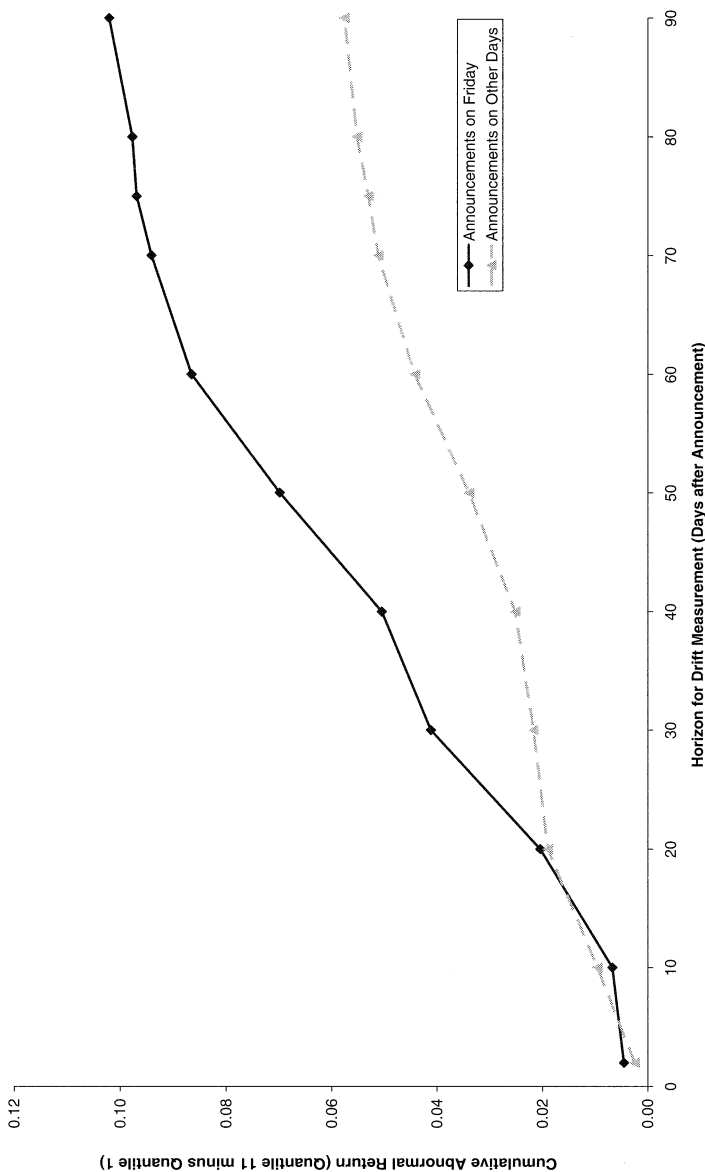


Figure 2. Performance of drift at different horizons. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year and announcements made on Saturday or Sunday are excluded from the sample. The measure for post-earnings announcement drift for horizon h is the average cumulative abnormal return from day 2 to day h for quantile 11 minus the average cumulative abnormal return from day 2 to day h for quantile 1.

and continues to increase during the next 60 trading days. These patterns suggest that the neglected information is not incorporated immediately following the weekend, but in a slow process. One possibility is that investors do not realize that they have ignored an announcement until they revisit their investment decisions for other reasons (e.g., subsequent news about the same company). Alternatively, investors might mistakenly believe that after missing the opportunity to trade on the day of or the day after the announcement there is no benefit to trading during the subsequent week. As Figure 2 shows, the timing of the increase in the differential drift is similar to the timing of the increase in drift for the baseline group (non-Friday announcements), which also appears largely after 30 trading days. This result is consistent with our preferred explanation that both drift and differential drift are due to the same underlying phenomenon, that is, the delayed response following inattention to information.

B.3. Long-Term Response

In Table II.C we measure the impact on long-term returns $R_{t,k}^{(0,75)}$. For the six specifications used in Table II.A and Table II.B, the top-bottom return differential on non-Fridays varies from 12.18% ($\hat{\phi}_{T-B} = 0.1218$, column 2) to 10.08% ($\hat{\phi}_{T-B} = 0.1008$, column 7). The top-to-bottom differential response on Fridays $\hat{\phi}_{T-B}^F$ is positive, significantly so in some of the specifications. The long-term response to earnings news thus is somewhat larger for Friday than for non-Friday announcements.

B.4. Summary

Stock prices respond less to Friday earnings surprises than to non-Friday earnings surprises immediately, event window (0,1). In the subsequent period, event window (2,75), stock prices respond *more* to Friday earning surprises, with smaller effects for the decile-based sorting procedure. Summing these two effects in the event window (0,75), since the drift result is larger in magnitude, Friday announcements have a somewhat larger long-term effect. These patterns are generally consistent with the predictions of the model if more investors are inattentive to the information released on Friday. For Friday announcements, inattention leads to less initial response, followed by more delayed response, as investors become aware of the neglected information.

C. Delayed Response Ratio

We implement a unified test of the model, suggested by Corollary 1, in Table II.D.⁷ We compute the share of the total stock response to announcements ($R_{t,k}^{(0,75)}$) that occurs with delay ($R_{t,k}^{(2,75)}$). We then test whether the delayed response ratio (DRR) is higher for Friday announcements. There are two advantages of this methodology: (i) it controls for heterogeneity in the long-term

⁷ We thank Owen Lamont for suggesting this approach.

reaction for Friday and non-Friday announcements (see Table II.C) by renormalizing with the long-term reaction; (ii) it offers an easily interpretable measure of the delay that provides a lower bound on the share of inattentive agents.

We compute the numerator and denominator of DRR as the difference in the average cumulative abnormal returns between the top and bottom quantile. The measure for non-Friday announcements is

$$DRR^{NF} = \frac{E[R_{k,t}^{(2,75)} | d_{t,k}^{top} = 1, d_{t,k}^F = 0] - E[R_{k,t}^{(2,75)} | d_{t,k}^{top} = 0, d_{t,k}^F = 0]}{E[R_{k,t}^{(0,75)} | d_{t,k}^{top} = 1, d_{t,k}^F = 0] - E[R_{k,t}^{(0,75)} | d_{t,k}^{top} = 0, d_{t,k}^F = 0]} = \frac{\phi_{T-B}^{(2,75)}}{\phi_{T-B}^{(0,75)}}, \tag{6}$$

where $\phi_{T-B}^{(2,75)}$ is the coefficient ϕ_{T-B} estimated in (5) with $R_{k,t}^{(2,75)}$ as dependent variable (and similarly for $\phi_{T-B}^{(0,75)}$) (Tables II.B and II.C). The standard errors are derived using the Delta method. In the specification without controls (column 1), 42.44% of the top-to-bottom stock response is delayed ($DRR^{NF} = 0.4244$). The results are remarkably similar with controls (columns 2 and 3),⁸ using deciles (column 4), and using the top two and bottom two quantiles (columns 5 to 7). For these different specifications, the average DRR lies between 0.40 and 0.45.

Similarly, we compute the delayed response ratio for Friday announcements:

$$DRR^F = \frac{E[R_{k,t}^{(2,75)} | d_{t,k}^{top} = 1, d_{t,k}^F = 1] - E[R_{k,t}^{(2,75)} | d_{t,k}^{top} = 0, d_{t,k}^F = 1]}{E[R_{k,t}^{(0,75)} | d_{t,k}^{top} = 1, d_{t,k}^F = 1] - E[R_{k,t}^{(0,75)} | d_{t,k}^{top} = 0, d_{t,k}^F = 1]} \tag{7}$$
$$= \frac{\phi_{T-B}^{(2,75)} + \phi_{T-B}^{F(2,75)}}{\phi_{T-B}^{(0,75)} + \phi_{T-B}^{F(2,75)}}.$$

In the benchmark specification (column 1), 61.87% of the top-to-bottom response on Friday occurs with delay ($DRR^F = 0.6287$). Compared to the delayed response ratio of 0.4244 on other weekdays, Fridays have an additional 19 percentage points of delayed reaction, an economically and statistically significant difference. The controls (columns 2 and 3) and the top two and bottom two quantiles (columns 5 and 6) have little effect on the results. In the specifications using deciles (columns 4 and 7), the difference is marginally significant in one specification and highly significant in the other. Hence, even in specifications for which the initial reaction or the delayed reaction is not significantly different (Tables II.A and II.B), there is a significant difference in the delayed response ratio for Friday and non-Friday announcements.

These results provide a lower bound for the share of distracted agents. Using Corollary 1, the estimates for DRR^{NF} and DRR^F imply that the share of distracted agents μ_t is larger than 0.4 for non-Friday announcements and larger

⁸ In the specifications with controls, we estimate the differential response ratio at the average value for each of the controls.

than 0.6 for Friday announcements. Quantitatively, inattention appears to be substantial.

To estimate whether the findings depend more on the response to positive announcements (top quantile) or negative announcements (bottom quantile), we compute the delayed response ratio separately for the top and bottom quantiles. In unreported results, the numerator and denominator of *DRR* are estimated using the difference between average returns in quantile j and average returns in quantile 6. Friday announcements have higher delayed response ratios for both quantile 1 and quantile 11, compared to non-Friday announcements.

C.1. Summary

For non-Friday announcements, 40% to 45% of the stock response is delayed. For Friday announcements, this figure is between 54% and 62%.⁹ According to the model, these figures provide a lower bound for the share of distracted agents. The substantially higher delayed response ratio for Friday announcements is consistent with increased distraction postponing investor response on Friday. This pattern is observable both for positive and for negative surprises, although the results are less precise for negative surprises.

D. Regressions Including All Announcements

In the regressions above, we restrict our attention to very positive and very negative earnings news. While this approach is simple and nonparametric, it does not take advantage of all available information. In order to use all the data, we impose identification restrictions on the relationship between earnings surprises and abnormal returns. Figure 1(a) indicates that the relationship between the earnings quantiles and the immediate stock response is approximately linear, with a differential slope for the Friday announcements. Linearity in the surprise quantile should not be confused with linearity in the surprise itself. As Figure 1(d) shows, the relationship between earnings surprises and the immediate stock response is monotonic but highly nonlinear, with a clear S-shape (Kothari (2001)). While the relationship between earnings surprise quantiles and the delayed stock response is more noisy, Figure 1(b) indicates that this relationship is approximately linear as well.

Following Hirshleifer et al. (forthcoming), we estimate the regression

$$R_{t,k}^{(h,H)} = \alpha + \beta S_{t,k} + \beta^F S_{t,k} d_{t,k}^F + \phi d_{t,k}^F + \Gamma_0 X_{t,k} + \Gamma_1 S_{t,k} X_{t,k} + \varepsilon_{t,k}, \quad (8)$$

where $S_{t,k}$ indicates the quantile corresponding to the earnings surprise and $X_{t,k}$ is the same set of controls that we use in Subsections B and C. The controls are present both in levels and interacted with the quantile function to allow for differential slopes of the function as a function of the controls. The coefficient β captures the average response of stock return for an increase of

⁹ The delayed response ratios for the Homogenous sample are similar to those for the whole sample.

one quantile in non-Friday announcements. The coefficient β^F captures the differential response of Friday earnings announcements, compared to non-Friday announcements. Following the model, this specification assumes that the response to Friday announcements is proportional to the response to non-Friday announcements conditional on the controls.

Columns 1 to 3 in Table III present the results of specification (8) using the short-run returns $R^{(0,1)}$ as dependent variable. The specification in column 1 without controls indicates that the short-run response for Friday announcement is 16.9% flatter than on other weekdays (0.0011/0.0065). The effect is precisely estimated and consistent with the finding of a 15.8% lower immediate response for the extreme quantiles (Table II.A). As we stated before, the assumption of linearity in the quantile embedded in this specification should not be confused with the assumption of linearity in the surprise itself. The R^2 of 0.0582 in the specification of column 1 is substantially higher than the R^2 of 0.0090 for a specification that is piece-wise linear in the surprise (instead of in the quantiles), $R_{t,k}^{(0,1)} = \alpha + \alpha_1 s_{t,k} + \phi^F s_{t,k} d_{t,k}^F + \gamma d_{t,k}^F + \varepsilon_{t,k}$.¹⁰ We obtain very similar results with the standard set of controls (column 2) and when we also control for earnings surprise volatility (not shown). The results are also similar when we use deciles instead of quantiles (column 3). Finally, when we allow for a different response on each weekday, the immediate stock response is significantly different (lower) only for Friday announcements (not shown).

In columns 4 to 6 in Table III we present the results of specification (8) using the long-term returns $R^{(2,75)}$ as the dependent variable. The specification in column 1 without controls indicates that the log-run response to Friday announcements is 56.2% larger than on other weekdays (0.0027/0.0048). This effect is also precisely estimated and consistent with the 77.2% increase for the extreme quantiles (Table II.B). The results are very similar using the standard set of controls (column 2), including additional controls for earnings surprise volatility (not shown), and replacing quantiles with deciles in specification (8) (column 3).

D.1. Summary

Friday announcements are associated with a 16.9% lower response of stock prices in the 2 days surrounding the announcement. These results are not sensitive to the introduction of controls or specification modifications, and are consistent with the results obtained using only the top and bottom quantile (Table II.A). We obtain a similarly robust result for the effect on the delayed stock response: It is 56.2% larger for Friday announcements.¹¹ The decreased

¹⁰ Bagnoli et al. (2005) use a piece-wise linear specification and find that the short-run response to Friday announcements is muted for negative surprises.

¹¹ In unreported regressions using the Homogeneous sample, there is less short-run attenuation and more differential drift for Friday announcements, compared to the whole sample.

Table III
Stock Price Response to Earnings Announcements for All Announcements

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are unequal, the number of observations in each quantile differ. In Columns 3 and 6 we follow a similar procedure but sort announcements into surprise deciles instead of quantiles. The breakpoints for the quantiles (deciles) are determined separately for each year. The standard set of controls includes indicators for the year of the announcement, the month of announcements, and the decile of a firm's market capitalization. The controls are present in levels and interacted with the quantile (decile) variable. Announcements made on Saturday or Sunday are excluded from the sample. Estimates from OLS regressions are reported in each column. Standard errors adjusted for heteroskedasticity and clustered by day of announcement are in parentheses (* significant at 10%; ** significant at 5%; *** significant at 1%).

	Dependent Variable: Cumulative Abnormal Return in Event Time from 0 to 1			Dependent Variable: Cumulative Abnormal Return in Event Time from 2 to 75		
	(1)	(2)	(3)	(4)	(5)	(6)
Friday	0.0067 (0.0023)***	0.0044 (0.0023)*	0.0025 (0.0019)	-0.0155 (0.0086)*	-0.0122 (0.0080)	-0.0098 (0.0069)
Earnings Surprise Quantile	0.0065 (0.0001)***	0.0047 (0.0004)***		0.0048 (0.0003)***	0.0069 (0.0015)***	
(Earnings Surprise Quantile) * Friday	-0.0011 (0.0003)***	-0.0008 (0.0003)***		0.0027 (0.0011)**	0.0023 (0.0011)**	
Earnings Surprise Decile			0.0051 (0.0004)***			0.0087 (0.0015)***
(Earnings Surprise Decile) * Friday			-0.0008 (0.0003)***			0.0024 (0.0011)**
Standard Controls (interacted)		X	X		X	X
R ²	0.0582	0.0619	0.0600	0.0028	0.0206	0.0205
N	N = 143,580	N = 143,580	N = 143,580	N = 143,580	N = 143,580	N = 143,580

initial response and larger subsequent response supports the hypothesis that Friday and the weekend temporarily distract investors.¹²

E. Portfolio Returns

The post-earnings announcement drift is substantially larger for Friday announcements than for non-Friday announcements. As an alternative measure of this differential drift, we present the returns to a portfolio that buys the Friday drift and sells the drift on other weekdays. This methodology mimics a feasible trading strategy designed to capture the specific type of inattention described in the model. It also demonstrates that the findings above are not induced by a pattern associated with a large number of announcements in a particular month or small number of months.

We analyze a monthly strategy from January 1995 to June 2006. The non-Friday drift portfolio for calendar month t purchases companies that, in month $t - 1$, made an announcement on a non-Friday in the top quantile; the portfolio sells short companies that, in month $t - 1$, made an announcement on a non-Friday in the bottom quantile.¹³ Therefore, the return for the non-Friday drift portfolio is $R_{NF}^D = R_{NF}^{11} - R_{NF}^1$, where the returns refer to calendar month t . The average non-Friday portfolio includes 168 stocks per month (113 long positions and 55 short positions). We compute monthly portfolio returns by equally weighting the relevant individual stock returns.¹⁴ Figure 3 shows year-by-year the average monthly return of this portfolio.

We construct a Friday drift portfolio for month t following a similar procedure except that we consider only companies that made an announcement on Friday during the previous month, $R_F^D = R_F^{11} - R_F^1$. The average Friday portfolio includes 15 stocks per month (eight long positions and seven short positions). The average monthly return for this portfolio is higher, approximately 4% per month (Figure 3), but more volatile given the smaller number of companies making Friday announcements. Nevertheless, in 10 years out of 12 the Friday drift portfolio has higher returns than the non-Friday drift portfolio, and it performs substantially worse only in 1 year.

In Table IV, we evaluate a monthly portfolio that purchases (long position) the Friday drift portfolio and sells (short position) the non-Friday drift portfolio, $R_{F-NF}^D = R_F^D - R_{NF}^D = R_F^{11} - R_F^1 - (R_{NF}^{11} - R_{NF}^1)$. We control for market performance by regressing the monthly differential drift on the CRSP

¹² We would like to test whether the decrease in the initial response occurs on Friday, before the weekend, or on Monday, after the weekend. However, this test requires that the fraction of announcements taking place after market close be similar for Friday and non-Friday announcements. Unfortunately, in recent years Friday announcements are significantly less likely to occur after market close according to Bagnoli et al. (2005), making a direct comparison impossible.

¹³ An announcement is part of the portfolio in month t if the second trading day after the announcement is in month $t - 1$. We compute the breakpoints for the 11 quantiles with the procedure described previously, except that we use the breakpoints from the prior calendar year. This approach guarantees that the trading strategy does not use any forward-looking information. If there are no stocks for a constituent portfolio, then that month is excluded from the performance analysis.

¹⁴ Value-weighting the returns within the constituent portfolios yields similar results.

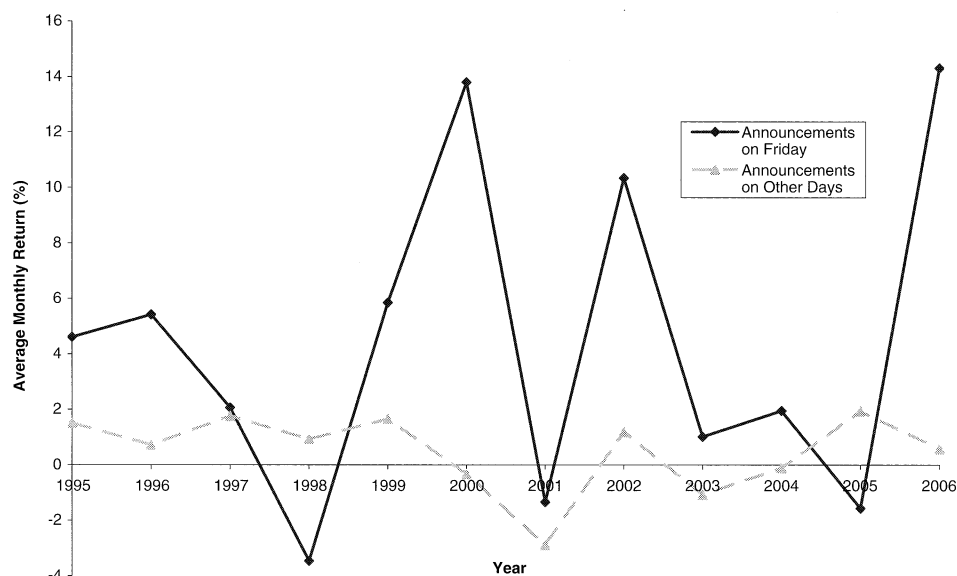


Figure 3. Average monthly return for zero-investment drift portfolios, yearly plot. For each year we report the average monthly return for the zero-investment portfolio for announcements on Friday and for announcements on other weekdays. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Announcements made on Saturday or Sunday are excluded from the sample. The sorting mechanism in the portfolio strategies analyzed below uses the breakpoints for the quantiles determined by non-Friday announcements during the previous calendar year. At the end of calendar month $t - 1$ the zero-investment drift portfolio for month t using Friday announcements (other weekdays) is created as follows. The strategy purchases stocks with earnings surprise announcements in the highest quantile and sells stocks with earnings surprises in the lowest quantile in month $t - 1$. For a stock to be included in the 1-month holding period strategy for calendar month t , the second trading day after the announcement must be in month $t - 1$. If there are no stocks in a constituent portfolio, then that month is excluded from the analysis. Returns are equally weighted within the constituent portfolios.

value-weighted stock index, net of the 1-month Treasury rate. The standard errors are corrected for heteroskedasticity and autocorrelation using the Newey-West estimator with six lags.¹⁵ The portfolio earns a significant monthly abnormal return of 3.84% (column 1). The estimated abnormal return increases somewhat if we include the size, book-to-market, and momentum factors (column 2). While the magnitude of this effect seems implausibly

¹⁵ The results do not change qualitatively if the lag length for the Newey-West estimator of the standard errors is 12.

Table IV
Performance of Post-Earnings Announcement Drift Portfolios

Stocks in CRSP are matched to quarterly earnings announcements in *I/B/E/S* from January 1995 until June 2006. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by *I/B/E/S* and the median analyst forecast included in the *I/B/E/S* detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quantiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quantiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Announcements made on Saturday or Sunday are excluded from the sample. The sorting mechanism in the portfolio strategies analyzed below uses the breakpoints for the quantiles determined by non-Friday announcements during the previous calendar year.

At the end of calendar month $t-1$ the zero-investment drift portfolio for month t using Friday announcements (other weekdays) is created as follows. The strategy purchases stocks with earnings surprises in the highest (or highest two) quantile(s) and sells stocks with earnings surprises in the lowest (or lowest two) quantile(s) in month $t-1$. The return for the differential drift portfolio is the difference between the return of the drift portfolio for Friday announcements and the return of the portfolio for non-Friday announcements. For a stock to be included in the 1-month holding period strategy for month t , the second trading day after the announcement must be in month $t-1$. For a stock to be included in the 2-month holding period drift strategy for month t , the second trading day after the announcement must be in month $t-1$ or in month $t-2$. If there are no stocks in a constituent portfolio, then that month is excluded from the analysis. Returns are equally weighted within the constituent portfolios.

Columns 1 through 6 report the coefficients of OLS regressions of the differential drift portfolio monthly returns from January 1995 to June 2006 on monthly factors. VWRP is the return on the CRSP value-weighted index minus the 1-month treasury rate. SMB, HML, and UMD are the returns on the factor-mimicking portfolios for size, book-to-market and momentum, respectively. The constant is the average monthly risk-adjusted return. Heteroskedasticity and autocorrelation consistent standard errors (in parentheses) are calculated using the Newey-West estimator with six lags (* significant at 10%, ** significant at 5%, *** significant at 1%). Columns 1 and 2 report results for the baseline specification. Column 3 only includes the non-Friday announcements that match the earnings surprises in the Friday drift portfolio (see details below). Column 4 uses announcements in the top two and bottom two quantiles. Column 5 analyzes a 2-month holding period. Column 6 replaces the quantile approach with a decile sorting procedure that is analogous to the baseline methodology where the breakpoints for the top and bottom groups are determined by sorting earnings surprises into deciles during the previous calendar year.

The portfolio strategy in column 3 (matched sample) is designed to address the concern that the stocks in the constituent Friday portfolios have more extreme earnings surprises than the stocks in the constituent non-Friday portfolios. For every stock with a Friday earnings announcement in the bottom (top) quantile for month $t-1$ we find the five non-Friday announcements in the same quantile and month with the closest earnings surprise to the Friday announcement subject to the constraint that non-Friday surprises must be larger in absolute value (more extreme). If there are fewer than five non-Friday announcements with more extreme earnings surprises than a specific Friday announcement, the returns of the stocks with these non-Friday announcements are overweighted appropriately. Friday announcements that can not be matched to at least one non-Friday announcement with a more extreme earnings surprise are excluded from the analysis. Only announcements in this matched sample are used for the analysis in column 3.

	Dependent Variable: Monthly Return on the Zero-Investment Portfolio					
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0384 (0.0134)***	0.0462 (0.0139)***	0.0584 (0.0220)***	0.0218 (0.0079)***	0.0232 (0.0086)***	0.0277 (0.0091)***
VW Index Excess Return (VWRF)	-0.2742 (0.3090)	-0.6419 (0.2778)**	-0.0968 (0.4262)	-0.1842 (0.1865)	-0.1068 (0.2301)	-0.4580 (0.1937)**

(continued)

Table IV—Continued

	Dependent Variable: Monthly Return on the Zero-Investment Portfolio					
	(1)	(2)	(3)	(4)	(5)	(6)
Size Factor Return (SMB)		0.2344 (0.4195)	0.5644 (0.6227)	-0.0390 (0.2464)	0.0701 (0.2930)	-0.0137 (0.2438)
Value Factor Return (HML)		-0.4607 (0.6143)	-1.5556 (0.7277)**	0.0762 (0.3329)	-0.3264 (0.2840)	-0.2094 (0.3820)
Momentum Factor Return (UMD)		-0.3994 (0.2632)	-1.1817 (0.6559)*	-0.0696 (0.1740)	-0.0410 (0.2206)	-0.3454 (0.1940)*
One month holding period	X	X	X	X	X	X
Two month holding period	X	X	X		X	
Top minus bottom quantile						
Matched sample			X			
Top two minus bottom two quantiles				X		
Top minus bottom decile						
R ²	0.0073	0.0385	0.1736	0.0152	0.0153	X 0.0398
N	N = 125	N = 125	N = 124	N = 130	N = 138	N = 127

large, we cannot reject a 1.8% monthly abnormal return, which is consistent with the estimated differential drift of 4.6 percentage points over 3 months (Table II.B).

One explanation of these results is that Friday announcements in the bottom and top quantiles have more extreme surprises, and that more extreme announcements display more drift. To address this explanation, we match announcements on the earnings surprise. For each Friday announcement in the bottom (top) quantile, we find the five non-Friday announcements in the same quantile and month with the closest earnings surprise, subject to the constraint that the non-Friday surprises must be larger in absolute value.¹⁶ The latter requirement guarantees that the differential drift results are not driven by the presence of more extreme news in the Friday portfolio. The portfolio formed according to this procedure (column 3) earns a monthly return of 5.84% that is somewhat larger than the return of the baseline portfolio (column 2).

The average abnormal return is smaller (but still significant) if we compute the portfolios using the top two quantiles and the bottom two quantiles (column 4). As in Table II.B, the differential Friday drift is smaller for these less extreme announcements. The portfolio return is lower (2.14%), albeit still significant, if we include companies that made announcements in months $t - 1$ or $t - 2$, rather than just in month $t - 1$ (column 5). Finally, in column 6 we determine the top and bottom groups by sorting the announcements into deciles instead of quantiles. The average return of the strategy, 2.77%, is significant; while the return is lower than for the quantile-based procedure (column 2), the significance level is higher. The lower average return is not surprising because the decile approach selects less extreme announcements in the bottom group.

E.1. Summary

We detect significantly higher portfolio returns from a drift strategy on Friday announcements than on non-Friday announcements. The differential return is generated largely in the first month after the announcement. The direction of the results is consistent with the findings in Table II.B, although the point estimates from the portfolio approach are larger.

IV. Volume Response

The immediate stock response to Friday earnings announcements is substantially lower than for non-Friday announcements. If this difference is caused by investor distraction, we expect a similar attenuation of trading volume in response to Friday news, because trading is the mechanism that causes prices to

¹⁶ If there are fewer than five non-Friday announcements with more extreme earnings surprises than a specific Friday announcement, the returns of the stocks with these non-Friday announcements are overweighted appropriately. Friday announcements that cannot be matched to at least one non-Friday announcement with a more extreme earnings surprise are excluded from the analysis.

adjust.¹⁷ If the difference is caused by higher dispersion of opinions for Friday announcements instead of inattention to Friday announcements, we expected greater abnormal volume rather than less volume after Friday announcements.

The measure for abnormal volume is

$$\Delta v_{t,k}^{(h,H)} = \sum_{u=\tau+h}^{\tau+H} \log(V_{t,k}^u) / (H - h + 1) - \sum_{u=\tau-20}^{\tau-11} \log(V_{t,k}^u) / 10, \quad (9)$$

where $V_{t,k}^u$ is the value of shares traded on day u and τ is the date of the earnings announcement in quarter t for company k . The measure $\Delta v_{t,k}^{(h,H)}$ is the percentage increase in volume around announcement date at horizon (h, H) for stock k in quarter t . In particular, $\Delta v_{t,k}^{(0,1)}$ is the immediate abnormal volume due to the announcement.

A. Graphical Evidence

Figure 4 plots the average abnormal volume $\Delta v_{t,k}^{(h,H)}$ at various horizons. For non-Friday announcements, the abnormal volume increases to 47% on the day of the announcement and then to 61% on the next trading day, before it decays slowly. A week after the announcement, the trading volume is still 13% higher than normal. The abnormal volume for Friday announcements is 13% higher compared to non-Friday announcements on the announcement day, and 43% lower on the next trading day. During the subsequent days, abnormal volume is slightly higher for non-Friday announcements.

The main difference between Friday and non-Friday announcements occurs during the day of and the day after the announcement. Abnormal volume $\Delta v_{t,k}^{(0,1)}$ is substantially lower for Friday than for non-Friday announcements.¹⁸ Our interpretation of this finding is that the weekend distracts investors. This interpretation also suggests why the difference between Friday and non-Friday announcements mostly disappears by day $\tau + 2$. Two offsetting forces are at work. On the one hand, some of the investors that have been distracted by the weekend are still not trading. On the other hand, other investors trade to respond to the underreaction after the weekend.

B. Regressions

We test whether the finding of lower abnormal volume $\Delta v_{t,k}^{(0,1)}$ for Friday announcements is affected by adding control variables. Specifically, we run the OLS specification

$$\Delta v_{t,k}^{(0,1)} = \alpha + \sum_{j \neq 6} \alpha_j d_{t,k}^j + \phi^F d_{t,k}^F + \Gamma X_{t,k} + \varepsilon_{t,k}. \quad (10)$$

¹⁷ This prediction does not follow from the model in Section I because the model has no natural definition of trading volume.

¹⁸ The measures $\Delta v_{t,k}^{(0,0)}$ and $\Delta v_{t,k}^{(1,1)}$ need to be considered together because the share of announcements after market close is lower on Friday.

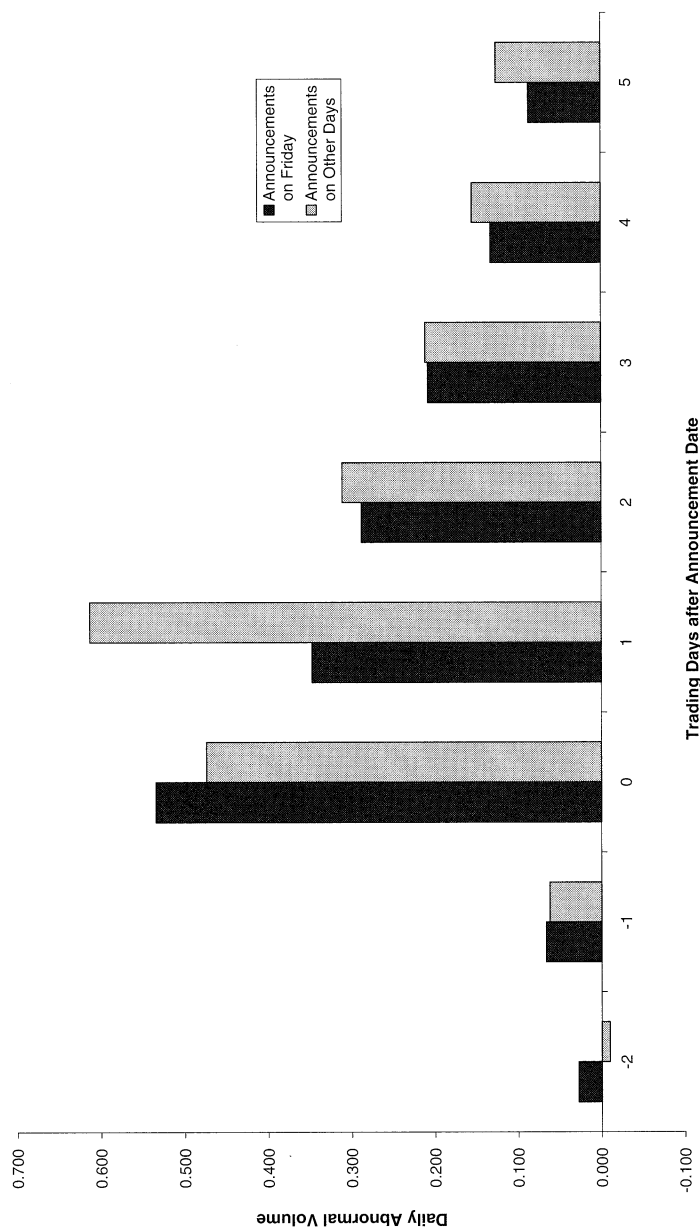


Figure 4. Abnormal volume around earnings announcement date. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The abnormal volume for each stock is the average log volume on the day of and the day after the announcement, divided by the average log volume for the period -20 to -11 in event time (10 trading days). Announcements made on Saturday or Sunday are excluded from the sample.

Table V
Short-Term Volume Response to an Earnings Announcement

Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The abnormal volume for each stock is the average log volume on the day of and the day after the announcement, divided by the average log volume for the period -20 to -11 in event time (10 trading days). The aggregate abnormal volume is computed in a similar manner using volume for the whole market. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement.

Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. Since the number of negative earnings surprises, positive earnings surprises, and surprises equal to zero are not equal, the number of observations in each quantile differ. In Column 5 we follow a similar procedure but sort announcements into surprise deciles instead of quantiles. The breakpoints for the quantiles (deciles) are determined separately for each year. Indicators for each earnings surprise quantile (decile) are included in all regressions.

The standard set of controls includes indicators for the year of announcement, the month of announcement, and the decile of a firm's market capitalization. The additional earnings surprise volatility controls in columns 4 and 5 are indicators for the decile of the company's earnings surprise standard deviation during the previous 4 years (requiring at least four observations). Announcements made on Saturday or Sunday are excluded from the sample. Estimates from OLS regressions are reported in each column. Standard errors adjusted for heteroskedasticity and clustered by day of announcement are in parentheses (* significant at 10%, ** significant at 5%, *** significant at 1%).

Dependent Variable: Abnormal Volume in Event Time from 0 to 1						
	(1)	(2)	(3)	(4)	(5)	(6)
Friday	-0.1109 (0.0161)***	-0.1235 (0.0130)***	-0.0436 (0.0129)***	-0.0413 (0.0130)***	-0.0414 (0.0129)***	-0.0173 (0.0121) 0.3977 (0.0289)***
Aggregate Abnormal Volume						
Earnings Surprise Quantiles	X	X	X	X		X
Earnings Surprise Deciles					X	
Standard Controls		X	X	X	X	X
Surprise Volatility Controls				X	X	
Company Fixed Effects			X	X	X	X
R ²	0.0072	0.0344	0.1963	0.1987	0.1984	0.2021
N	N = 143,369	N = 143,369	N = 143,369	N = 122,325	N = 122,325	N = 143,369

The terms $d_{t,k}^j$ are indicators for earning surprises $s_{t,k}$ in the j -th quantile, with quantile 6 ($s_{k,t} = 0$) as the omitted category. The control variables $X_{t,k}$ are month, year, and size indicators, as well as earnings surprise volatility indicators in one specification. Without controls (column 1 of Table V), the Friday coefficient is negative and significant, $\hat{\phi}^F = -0.1109$. Compared to the average immediate volume increase on non-Fridays, .5416, the abnormal volume increase on Friday is 20.5% lower ($0.1109/0.5416 = 0.2047$). With controls (column 2), the Friday effect is larger ($\hat{\phi}^F = -0.1235$).

In column 3 we introduce firm fixed effects to control for any company-level difference in abnormal volume. Some of the Friday effect appears to be due to heterogeneity in abnormal volume among firms, but the remaining effect ($\hat{\phi}^F = -0.0436$) is still large and significant. The 8.1% attenuation of abnormal volume for Friday announcements ($0.0436/0.5416 = 0.0081$) is of the same

order of magnitude as the 15% attenuation of the immediate stock return reaction (Tables II.A and III). We include firm fixed effects in the subsequent specifications. The results are very similar after controlling for earnings surprise volatility (column 4), using decile indicators instead of quantile indicators (column 5), and in the Homogeneous sample (not shown).

The attenuation of abnormal volume for Friday announcements could be explained by an unobserved variable that leads to lower aggregate market volume on Friday and Monday. To test for this explanation, we include abnormal aggregate volume as a control.¹⁹ After introducing this control, the Friday effect is smaller and not statistically significant (column 6). While this result may be a symptom of an omitted variable, it is also consistent with the inattention hypothesis. Inattention predicts that volume should be lower in response to any information on Friday and Monday, and hence aggregate volume should also be lower.

C. Summary

Short-term abnormal volume is significantly lower for Friday earnings announcements, even after controlling for announcement quality, control variables, and firm-specific variation. The Friday effect in abnormal volume is of the same order of magnitude as the Friday effect in immediate returns. The effect is not statistically significant after controlling for aggregate volume. The volume results are consistent with the hypothesis that investors underreact initially to information released on Friday.

V. Alternative Explanations

In this section we consider four alternative explanations of the findings, information processing time, pre-announcement release, announcement delay, and firm heterogeneity. We also discuss an attention-related interpretation that is inconsistent with the data.

A. Information Processing Time

The mechanics of Friday and non-Friday announcements differ. First, Friday announcements during the 1970s were more likely to be made when the market was closed (Patell and Wolfson (1982)). Second, Friday announcements are followed by a 2-day break in trading. For both reasons, investors have a different amount of time to process Friday announcements. However, these factors bias the analysis against finding underreaction. During our sample period, Friday announcements are *less* likely to be released after the market is closed (Bagnoli et al. (2005)). (This explains why more of the volume response for Friday

¹⁹ We measure aggregate volume as the equal-weighted average across firms of the number of shares traded, divided by shares outstanding. We generate the abnormal aggregate volume $\Delta v_{t,A}^{(0,1)}$ using equation (9).

announcements occurs on the same day instead of the next day—Figure 4). Moreover, the 2-day break in trading allows investors more time (not less) to assimilate new information.

B. Pre-announcement Release

Companies release the date of the earnings announcements in advance. In the event of poor performance, companies may also issue earnings warnings. These pre-announcement releases could explain the lower immediate reaction of stock prices to Friday announcements. Assume that firms announcing on Friday are more likely to issue earnings warnings. Additionally, the decision to announce on Friday, itself, could be interpreted as a warning about earnings. Stock prices may respond before the official earnings announcement. Because forecasts are not always revised after these pre-announcements, the negative surprises constructed from the media forecast may overestimate the “true” surprise to investors. This hypothesis can explain a lower short-term reaction of stock returns to negative surprises. However, it does not explain the attenuated short-term reaction for positive surprises. If anything, it predicts a stronger short-term response to positive announcements. This interpretation is also unable to explain differential drift. Further, we can directly test whether investors perceive more negative news before Friday announcements, conditional on the quantile of the surprise. Figure 1(c) displays stock returns for the horizon $(-30, -1)$ as a function of the quantiles of earnings surprises. In each of the outermost quantiles, the average pre-announcement returns are very similar for Friday and non-Friday announcements.

C. Announcement Delay

Stock prices decline before an earnings announcement if the announcement is delayed (Begley and Fischer (1998), Bagnoli, Kross, and Watts (2002)). It is possible that Friday announcements are more likely to be late. Hence, part of the stock price reaction may occur before the earnings announcement. However, Bagnoli et al. (2002) also find that the response to earnings information is similar for on-time and late announcements. Late announcements, therefore, do not explain the flatter short-term response to Friday earnings announcements. In addition, delaying an announcement is essentially a pre-announcement warning. As we discussed above, there is no evidence of lower pre-announcement returns for negative earnings surprises announced on Friday (relative to other weekdays), after controlling for the earnings surprise quantiles (Figure 1(c)).

D. Firm Heterogeneity

The attenuated immediate response on Friday could be due to unobserved heterogeneity. For example, the news about future profitability embedded in earnings announcements may be more transitory for firms making Friday

announcements, or may reflect more divergence in opinions. Some of these explanations are not consistent with other findings. For example, if the Friday news were more temporary, we would expect less drift; if they reflected a higher divergence in opinions, we would expect a higher initial abnormal volume. While it is impossible to fully control for all forms of heterogeneity, we show that the results remain qualitatively unchanged after the introduction of time, company, and market capitalization controls.

E. Task Overload

An attention-related interpretation is that investors on Monday are overburdened by information that has accumulated on their desks during the weekend and they find less time to react to Friday earnings announcements. However, aggregate volume is 10% lower on Monday, contrary to the hypothesis that traders are overwhelmed by this information.

VI. Conclusion

We compare the reaction to earnings announcements on Friday to the reaction on other weekdays. Friday announcements are characterized by a lower immediate response and a higher delayed response. The delayed response as a percentage of the total response is 60% on Friday and 40% on other weekdays. A portfolio that invests in the differential Friday drift earns substantial returns. We observe analogous results for volume.

The evidence supports the inattention hypothesis. On Friday, investors are distracted from work-related activities. Given limited attention, distractions cause underreaction to the earnings information. Eventually, investors recognize the mispricing and incorporate the information. Since a proxy for inattention increases the delayed reaction to earnings announcements, our findings indicate that underreaction to new information is an important source of post-earnings announcement drift.

Appendix

A. Equilibrium

In equilibrium, total demand must equal total supply for the risky asset or $\int_0^1 \lambda_t^i di = 1$. After substituting for λ_t^i in this expression, we solve for P_t :

$$\begin{aligned} P_t &= \frac{b_t E_t^\mu [D_{t+1} + P_{t+1}] + (1 - b_t) E_t^{1-\mu} [D_{t+1} + P_{t+1}] - a_t}{1 + R} \\ &= \frac{\bar{E}_t [D_{t+1}] + \bar{E}_t [P_{t+1}] - a_t}{1 + R}, \end{aligned} \quad (\text{A1})$$

where $a_t = (\frac{\mu_t}{\gamma\sigma_{\mu,t,P+D}^2} + \frac{1-\mu_t}{\gamma\sigma_{1-\mu,t,P+D}^2})^{-1}$, $\bar{E}_t = b_t E_t^\mu[\cdot] + (1-b_t)E_t^{1-\mu}[\cdot]$, and $b_t = \frac{\mu_t}{\gamma\sigma_{\mu,t,P+D}^2} a_t$. Recursively solving forward expression and also imposing the relevant transversality condition $\lim_{T \rightarrow \infty} (\bar{E}_t \bar{E}_{t+1} \dots \bar{E}_{t+T+1} [P_{t+T+2}]) / (1+R)^{T+2} = 0$ yields

$$P_t = \frac{\bar{E}_t [D_{t+1}] - a_t}{1+R} + \sum_{s=1}^{\infty} \frac{\bar{E}_t \bar{E}_{t+1} \dots \bar{E}_{t+s} [D_{t+s+1} - a_{t+s}]}{(1+R)^{s+1}}.$$

Since the information available at time t is uninformative regarding D_{t+s+1} , $\forall s > 0$, we know that $\bar{E}_{t+1} \bar{E}_{t+2} \dots \bar{E}_{t+s} [D_{t+s+1}] = \delta$. Since future signals about dividends are not forecastable, the strategic release of future signals is unforecastable as well. As long as perceived variances are time-invariant (verified below), $E_{t+s-1}^\mu [a_{t+s}] = E_{t+s-1}^{1-\mu} [a_{t+s}] = \bar{a}$, where $\bar{a} = \eta \alpha(\mu_h) + (1-\eta) \alpha(\mu_l)$ and $\eta = \text{prob}(\mu_{t+s} = \mu_h)$. Consequently, $\bar{E}_t \bar{E}_{t+1} \dots \bar{E}_{t+s} [a_{t+s}] = \bar{a}$ and the formula for the equilibrium price follows.

B. Time-Invariant Perceived Variances

We show that $\sigma_{\mu,t,P+D}^2$ and $\sigma_{1-\mu,t,P+D}^2$ do not depend on t . First, we solve for $\sigma_{\mu,t,P+D}^2$:

$$\begin{aligned} \sigma_{\mu,t,P+D}^2 &= E_t^\mu [(P_{t+1} + D_{t+1} - E_t^\mu [P_{t+1} + D_{t+1}])^2] \\ &= E_t^\mu \left[\left(\frac{(1-b_{t+1})s_{t+1} - a_{t+1} - (E_t^\mu [(1-b_{t+1})s_{t+1}] - \bar{a})}{1+R} + s_t + \varepsilon_{t+1} \right)^2 \right] \\ &= \left(\frac{1}{1+R} \right)^2 \sigma_{(1-b)s-a}^2 + \sigma_s^2 + \sigma_\varepsilon^2. \end{aligned}$$

Similarly, $\sigma_{1-\mu,t,P+D}^2 = (\frac{1}{1+R})^2 \sigma_{(1-b)s-a}^2 + \sigma_s^2$. If the strategic behavior of managers is time-invariant (Proposition 2), neither perceived variance expression depends on t .

C. Comparative Statics for a_t and b_t

Let $k = (1+R)^{-2} \sigma_{(1-b)s-a}^2 + \sigma_s^2$. We substitute the expressions for $\sigma_{\mu,P+D}^2$, $\sigma_{1-\mu,P+D}^2$, and k into the definition of a_t and rearrange to obtain

$$a_t = \gamma(k^2 + k\sigma_s^2)(k + \sigma_s^2(1-\mu_t))^{-1}.$$

Since $k > 0$, it follows that $a_h > a_l$. Using a similar procedure,

$$b_t = k\mu_t(k + \sigma_s^2(1-\mu_t))^{-1},$$

which implies that $b_h > b_l$ and $b_t(\mu_t = 0) = 0$.

Proof of Proposition 1: For part (i) we start with the definition of IR_t ,

$$\begin{aligned} E_t[IR_t] &= E_t[Z_t - E_{t-1}[Z_t] - (D_t - E_{t-1}[D_t])] = P_t - E_{t-1}[P_t] \\ &= -\frac{a_t - \bar{a}}{1+R} + \frac{(1-b_t)s_t - E_{t-1}[(1-b_t)s_t]}{1+R} \\ &= \frac{\bar{a} - a_t}{1+R} + \frac{\sigma_{bs}}{1+R} + \frac{1-b_t}{1+R}s_t. \end{aligned}$$

The remainder of the proof follows from $b_h > b_l$ and $b_t(\mu_t = 0) = 0$. For part (ii) we start with the definition of $E_t[Z_{t+1}]$:

$$\begin{aligned} E_t[Z_{t+1}] &= E_t[P_{t+1} + D_{t+1} - (1+R)P_t] \\ &= -\frac{E_t[b_{t+1}s_{t+1}]}{1+R} + s_t + a_t - (1-b_t)s_t = a_t - \frac{\sigma_{bs}}{1+R} + b_t s_t. \end{aligned}$$

This implies $E_t[DR_t] = E_t[Z_{t+1}] - E_{t-1}[Z_{t+1}] = a_t - \bar{a} + b_t s_t - E_{t-1}[b_t s_t] = a_t - \bar{a} - \sigma_{bs} + b_t s_t$. The remainder of the proof follows from $b_h > b_l$ and $b_t(\mu_t = 0) = 0$. For part (iii) we rewrite that $Z_{t+1,t-1}$ as $Z_{t+1} + (1+R)Z_t$. We substitute this expression for $Z_{t+1,t-1}$ into the expression for $E_t[LR_{t+1}]$ (in the text) to obtain

$$\begin{aligned} E_t[LR_{t+1}] &= [(E_t[Z_t] - E_{t-1}[Z_{t-1}]) - (E_t[D_t] - E_{t-1}[D_t])] \\ &\quad + \frac{E_t[Z_{t+1}] - E_{t-1}[Z_{t+1}]}{1+R} \\ &= E_t[IR_t] + \frac{E_t[DR_t]}{1+R}. \end{aligned}$$

The remainder of the proof follows from this expression for $E_t[LR_{t+1}]$ and parts (i) and (ii). Q.E.D.

Proof of Corollary 1: Part (i) follows from the expressions for $E_t[DR_t]$ and $E_t[LR_{t+1}]$ in the proof of Proposition 1. Part (ii) follows from $b_h > b_l$ and $b_t(\mu_t = 0) = 0$. Part (iii) follows from $R > 0$ and the definition of b_t because $\sigma_{1-\mu,t,P+D}^2$ is necessarily smaller than $\sigma_{\mu,t,P+D}^2$. Q.E.D.

Proof of Proposition 2: Part (i) is discussed in the body of the paper. We prove part (ii). The problem of the short-term manager is equivalent to solving $\max_{\mu_t \in \{\mu_h, \mu_l\}} -a_t - b_t s_t$. Substituting in the values of a_t and b_t , we obtain

$$\max_{\mu_t \in \{\mu_h, \mu_l\}} \left(-\gamma - \frac{\mu_t s_t}{k + \sigma_s^2} \right) \left(\frac{\mu_t}{k + \sigma_s^2} + \frac{1 - \mu_t}{k} \right)^{-1},$$

where $k = (1+R)^{-2}\sigma_{bs-a}^2 + \sigma_\varepsilon^2 > 0$. Announcing on a high distraction day is optimal if and only if

$$\left(-\gamma - \frac{\mu_h s_t}{k + \sigma_s^2} \right) \left(\frac{\mu_h}{k + \sigma_s^2} + \frac{1 - \mu_h}{k} \right)^{-1} \geq \left(-\gamma - \frac{\mu_l s_t}{k + \sigma_s^2} \right) \left(\frac{\mu_l}{k + \sigma_s^2} + \frac{1 - \mu_l}{k} \right)^{-1}.$$

Table A.I
Accuracy of the Imputed Date for Earnings Announcements

Newswire searches using Lexis-Nexis provide the actual date for the earnings announcement. The imputed date for the earnings announcement is generated from the recorded announcement dates in Compustat and I/B/E/S using a simple algorithm. The algorithm is described in the text and is designed to maximize the match between imputed and actual dates in the sample of 2,487 observations randomly selected for a newswire search.

Difference in Trading Days	Imputed Announcement Date		Same Date	Imputed Announcement Date		
	Before Actual (Newswire) Date	After Actual (Newswire) Date		1	2	Other
	-2	-1	0			Total
Panel A: Years 1984 to 1989						
Number of Imputed Friday Announcements	0	47	89	50	2	4
	0.0%	24.5%	46.4%	26.0%	1.0%	2.1%
Number of Imputed Non-Friday Announcements	2	87	322	58	7	8
	0.4%	18.0%	66.5%	12.0%	1.4%	1.7%
						100.0%
Panel B: Years 1990 to 1994						
Number of Imputed Friday Announcements	0	4	187	96	4	1
	0.0%	1.4%	64.0%	32.9%	1.4%	0.3%
Number of Imputed Non-Friday Announcements	1	3	310	19	1	1
	0.3%	0.9%	92.5%	5.7%	0.3%	0.3%
						100.0%
Panel C: Years 1995 to 2002						
Number of Imputed Friday Announcements	0	17	609	11	0	1
	0.0%	2.7%	95.5%	1.7%	0.0%	0.2%
Number of Imputed Non-Friday Announcements	1	13	527	3	0	2
	0.2%	2.4%	96.5%	0.5%	0.0%	0.4%
						100.0%

Multiplying the terms together and simplifying leads to $\mu_h s_t - \gamma \sigma_s^2 \mu_l \leq \mu_l s_t - \gamma \sigma_s^2 \mu_h$, and finally to $s_t \leq -\sigma_s^2 \gamma$, the desired condition. Given this threshold rule, $E[s_t | \mu_t = \mu_h] < 0$ and $E[s_t | \mu_t = \mu_l] > 0$ follow immediately. Hence, $E[s_t | \mu_t = \mu_h] < E[s_t | \mu_t = \mu_l]$. The relationship $E[IR_t | \mu_t = \mu_h] < E[IR_t | \mu_t = \mu_l]$ follows from

$$\begin{aligned} & E[IR_t | \mu_t = \mu_h] - E[IR_t | \mu_t = \mu_l] \\ &= \frac{a_l - a_h}{1 + R} + \frac{1 - b_h}{1 + R} E[s_t | \mu_t = \mu_h] - \frac{1 - b_l}{1 + R} E[s_t | \mu_t = \mu_l] \end{aligned}$$

because $a_l - a_h < 0$, $E[s_t | \mu_t = \mu_h] < 0$, and $E[s_t | \mu_t = \mu_l] > 0$. Q.E.D.

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